

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO3_AT1 — EXPERIMENT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Construction (BL3)	Assessment:	Assessment Tool 1 – Experiment
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data.(BL3)</i>		
Student Name:	JOTHIKA M	Reg. No.:	192324235
Section / Batch:	SLOT C	Date:	11/08/2026

Code of Conduct

I, JOTHIKA M (1912324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

Annexure A – EXPERIMENT

Problem 2 (5 Marks)

Data: Monthly household budget allocation: Rent = 40%, Food = 25%, Transport = 15%, Savings = 20%.

Task: Construct a pie chart representing this budget breakdown, labeling each slice with its percentage.

Problem 9 (5 Marks)

Data: Monthly salaries (in ₹1,000s) of 8 employees: 25, 28, 30, 30, 30, 45, 48, 50.

Task: Construct a bean plot and identify any clusters or repeated values visible in the plot.

Problem 11 (5 Marks)

Data: Exam scores of 12 students: 45, 50, 55, 58, 60, 62, 63, 65, 68, 70, 75, 85.

Task: Construct a violin plot for this data and describe where the distribution is widest (most concentrated).

Problem 18 (5 Marks)

Data: Test scores for two sections — Section A: 60, 65, 68, 70, 72, 75, 80. Section B: 40, 50, 55, 60, 65, 70, 95.

Task: Construct box plots for both sections and compare their medians and spread.

Problem 25 (5 Marks)

Data: Household budget by category across 2 months: Month 1 — Rent = 15,000, Food = 8,000, Other = 5,000. Month 2 — Rent = 15,000, Food = 9,000, Other = 6,000.

Task: Construct a stacked bar chart showing total spending and category composition for both month.

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Experiment

CO3 · BT3: Apply ·

This rubric is used to assess student responses to the CO3-AT1 Experiment, in which students are given a small dataset and must construct an accurate pie chart, bean plot, violin plot, box plot, or stacked bar chart.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of the correct chart type (pie, bean, violin, box, or stacked bar) and the elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type or required elements.	Poor understanding of core construction concepts; selects a chart type fundamentally unsuited to the data.
Explanation	25%	Shows detailed, clearly worked calculations (e.g., pie chart angles, five-number summary, IQR, stacked totals) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	20%	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.
Organization	15%	The chart is neatly and logically constructed, with a title, correctly labeled axes or slices, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, labels, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, labels, or a legend.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Accuracy	10%	The chart is completely accurate — all values, proportions, quartiles, and calculations correctly match the given data.	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2 – Pie Chart – Monthly Household Budget Allocation

Introduction

A pie chart is a circular visualization used to represent how a total quantity is divided among different categories. Each category is represented by a slice of the circle, and the size of the slice is proportional to its percentage of the total budget.

In this problem, the monthly household budget is divided into **Rent, Food, Transport, and Savings**. The total allocation is 100%, so the four categories together form a complete circle of **360°**.

Given Data

Category	Budget Allocation
Rent	40%
Food	25%
Transport	15%
Savings	20%
Total	100%

Angle Calculation

The angle of each slice can be calculated using:

$$\text{Angle} = \frac{\text{Percentage}}{100} \times 360^\circ$$

Rent

$$\frac{40}{100} \times 360 = 144^\circ$$

Food

$$\frac{25}{100} \times 360 = 90^\circ$$

Transport

$$\frac{15}{100} \times 360 = 54^\circ$$

Savings

$$\frac{20}{100} \times 360 = 72^\circ$$

Therefore:

$$144+90+54+72=360^\circ$$

R Code

```
R
# Install package if required
# install.packages("ggplot2")

library(ggplot2)

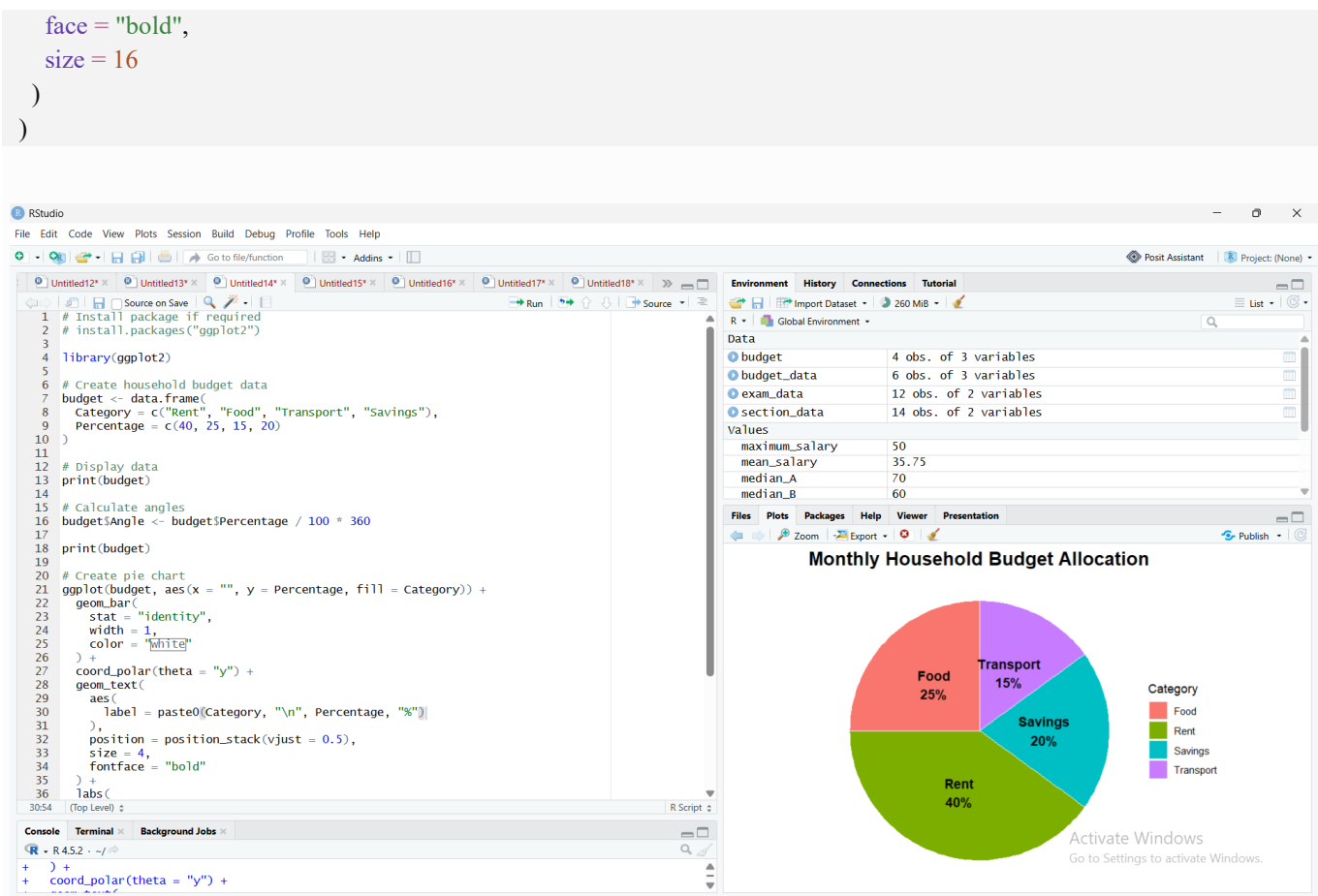
# Create household budget data
budget <- data.frame(
  Category = c("Rent", "Food", "Transport", "Savings"),
  Percentage = c(40, 25, 15, 20)
)

# Display data
print(budget)

# Calculate angles
budget$Angle <- budget$Percentage / 100 * 360

print(budget)

# Create pie chart
ggplot(budget, aes(x = "", y = Percentage, fill = Category)) +
  geom_bar(
    stat = "identity",
    width = 1,
    color = "white"
  ) +
  coord_polar(theta = "y") +
  geom_text(
    aes(
      label = paste0(Category, "\n", Percentage, "%")
    ),
    position = position_stack(vjust = 0.5),
    size = 4,
    fontface = "bold"
  ) +
  labs(
    title = "Monthly Household Budget Allocation",
    fill = "Category"
  ) +
  theme_void() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
```



Evidence

The pie chart clearly shows that **Rent occupies the largest portion of the household budget at 40%**. Food accounts for 25%, Savings account for 20%, and Transport represents the smallest allocation at 15%.

The corresponding angles are:

- Rent → 144°
- Food → 90°
- Transport → 54°
- Savings → 72°

Interpretation

The household allocates a significant portion of its income toward housing expenses. Rent consumes 40% of the budget, making it the dominant expense. Food is the second-largest expense at 25%.

Savings account for 20%, which indicates that a considerable portion of the budget is reserved for future financial needs. Transport receives the smallest allocation at 15%.

The chart makes it easy to identify which categories consume the greatest share of the household budget.

Conclusion

The pie chart provides a simple and effective representation of household budget allocation. It clearly shows the relative contribution of each category and allows the viewer to quickly identify **Rent as the largest expense and Transport as the smallest expense.**

Problem 9 – Bean Plot – Monthly Salaries

Introduction

A bean plot is a statistical visualization used to show the **distribution of numerical data**. It combines features of a density plot and individual observations. The shape of the bean indicates where observations are concentrated, while individual points help identify repeated or unusual values. In this problem, monthly salaries of eight employees are analyzed to identify clusters and repeated salary values.

Given Data

Monthly salaries are given in ₹1,000s:

25, 28, 30, 30, 30, 45, 48, 50

Employee	Salary (₹1,000s)
Employee 1	25
Employee 2	28
Employee 3	30
Employee 4	30
Employee 5	30
Employee 6	45
Employee 7	48
Employee 8	50

Basic Analysis

Minimum salary

25,000

Maximum salary

50,000

Salary range

$50 - 25 = 25$

Therefore, the salary range is **₹25,000**.

Repeated value

The salary **₹30,000** occurs **three times**.

R Code

```
R
# Install package if required
# install.packages("beanplot")

library(beanplot)

# Salary data
salary <- c(
  25, 28, 30, 30,
  30, 45, 48, 50
)

# Calculate basic statistics
mean_salary <- mean(salary)
```



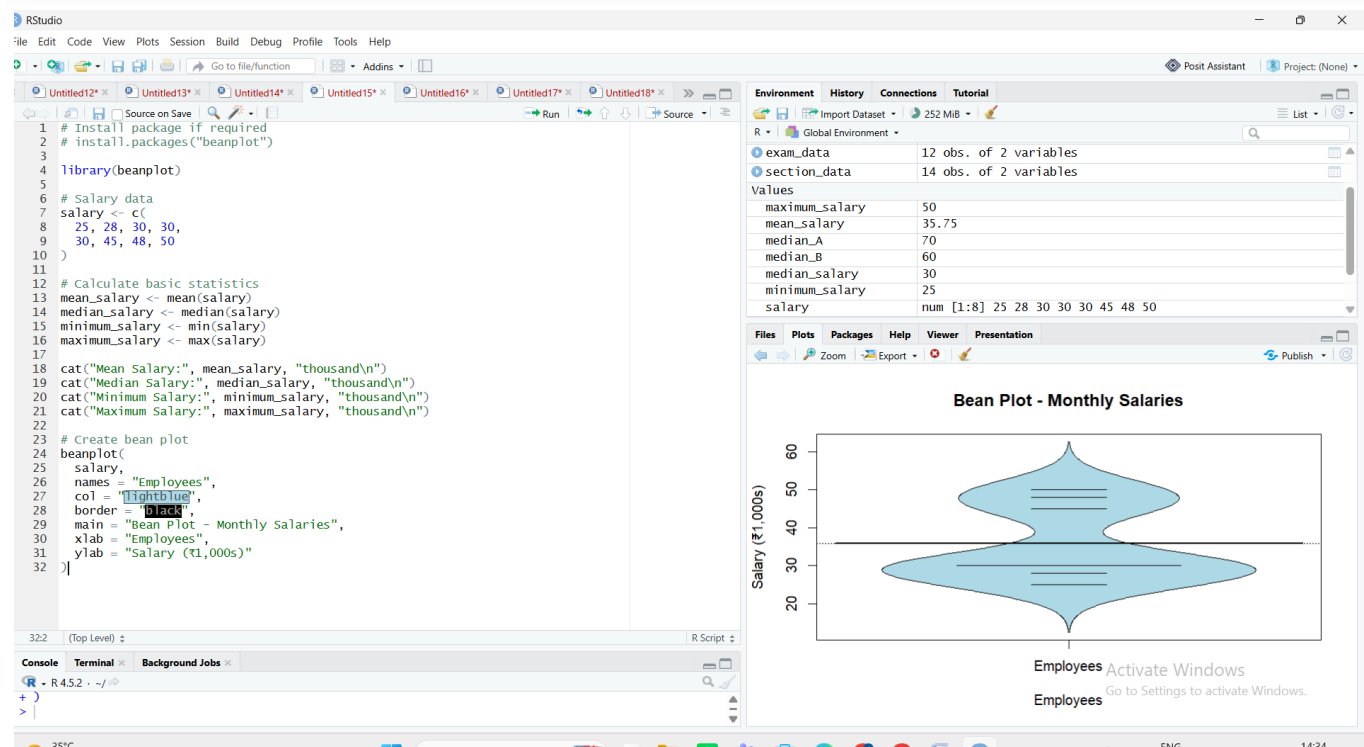
```

median_salary <- median(salary)
minimum_salary <- min(salary)
maximum_salary <- max(salary)

cat("Mean Salary:", mean_salary, "thousand\n")
cat("Median Salary:", median_salary, "thousand\n")
cat("Minimum Salary:", minimum_salary, "thousand\n")
cat("Maximum Salary:", maximum_salary, "thousand\n")

# Create bean plot
beanplot(
  salary,
  names = "Employees",
  col = "lightblue",
  border = "black",
  main = "Bean Plot - Monthly Salaries",
  xlab = "Employees",
  ylab = "Salary (₹1,000s)"
)

```



Evidence

The salaries range from **₹25,000 to ₹50,000**.

The strongest repeated value is:

₹30,000 → 3 employees

The data can also be viewed as two broad groups:

- Lower salary group → ₹25,000–₹30,000
- Higher salary group → ₹45,000–₹50,000

There is a noticeable gap between ₹30,000 and ₹45,000.

Interpretation

The bean plot reveals a concentration around **₹30,000**, because this salary occurs three times. The remaining salaries are distributed toward the higher range.

The distribution therefore suggests that employees are not evenly distributed across the salary range. A small lower-salary cluster occurs around ₹25,000–₹30,000, while another group is visible around ₹45,000–₹50,000.

The repeated value of ₹30,000 is particularly important because it represents the most common salary in the dataset.

Conclusion

The bean plot effectively displays both the **distribution and repeated observations** in the salary data. The main cluster is around **₹30,000**, with additional observations at higher salary levels.

Problem 11 – Violin Plot – Exam Scores

Introduction

A violin plot is a statistical visualization that combines the characteristics of a **box plot and a density plot**. It displays the distribution, spread, central tendency, and concentration of numerical data.

The width of a violin at a particular value represents the **density of observations** at that value. A wider section indicates greater concentration, while a narrower section indicates fewer observations.

Given Data

Exam scores:

45, 50, 55, 58, 60, 62, 63, 65, 68, 70, 75, 85

Student	Score
1	45
2	50
3	55
4	58
5	60
6	62
7	63
8	65
9	68
10	70
11	75
12	85

Basic Analysis

Minimum:

45

Maximum:

85

Range:

$$85 - 45 = 40$$

The majority of scores occur between approximately **55 and 75**.

R Code

```
R
# Install package if required
# install.packages("ggplot2")

library(ggplot2)

# Exam score data
scores <- c(
  45, 50, 55, 58,
  60, 62, 63, 65,
  68, 70, 75, 85
)

exam_data <- data.frame(
  Group = "Students",
  Score = scores
)

# Display basic statistics
cat("Mean:", mean(scores), "\n")
cat("Median:", median(scores), "\n")
cat("Minimum:", min(scores), "\n")
cat("Maximum:", max(scores), "\n")
cat("Range:", max(scores) - min(scores), "\n")

# Violin plot
ggplot(exam_data, aes(x = Group, y = Score)) +

  geom_violin(
    fill = "skyblue",
    color = "black",
    alpha = 0.7,
    trim = FALSE
  ) +

  geom_boxplot(
    width = 0.12,
    fill = "white",
    color = "black"
  ) +

  geom_jitter(
    width = 0.05,
    color = "red",
    size = 2
  ) +

  labs(
    title = "Violin Plot - Student Exam Scores",
    x = "Group",
```

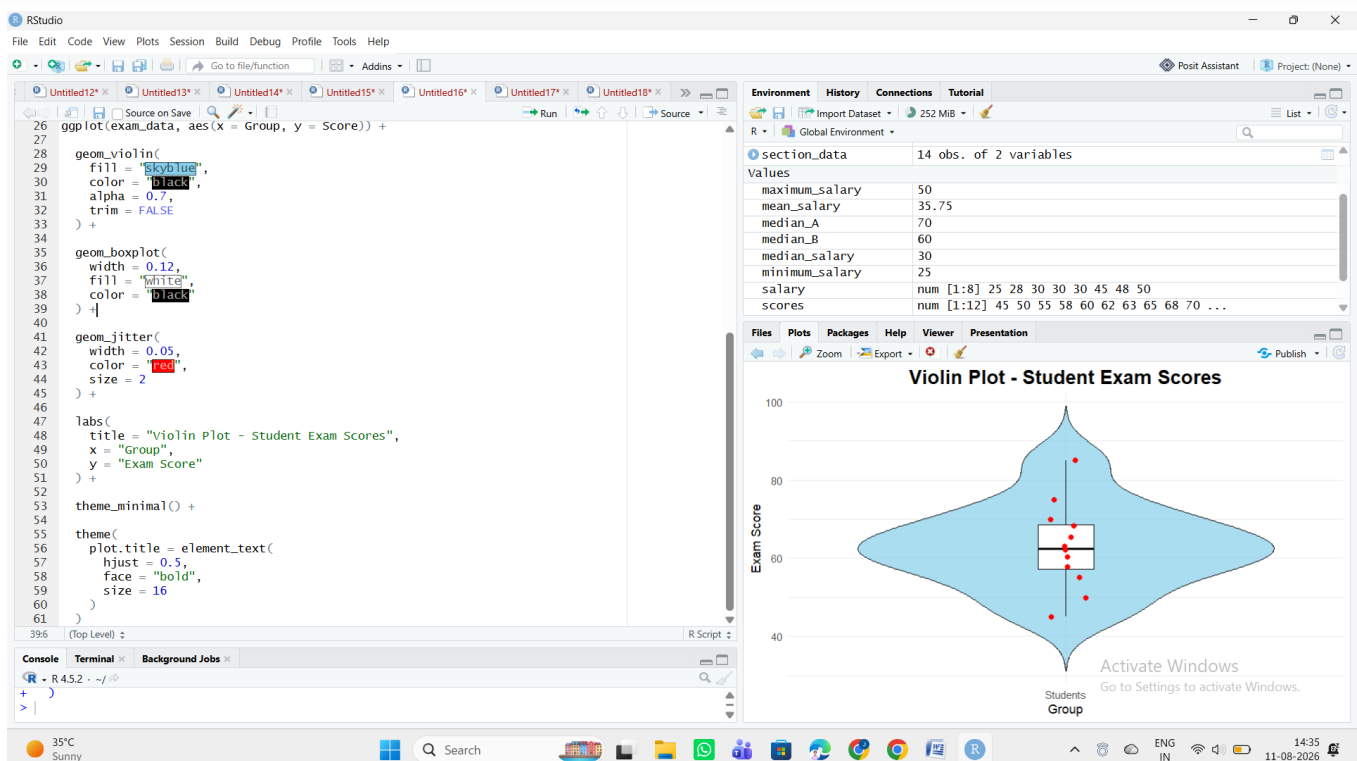
```

y = "Exam Score"
) +

theme_minimal() +

theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 16
  )
)

```



Evidence

The exam scores range from **45 to 85 marks**.

Most observations lie between **55 and 75 marks**. The scores around **60–70** form the central concentration of the data.

The score of 85 is at the upper extreme compared with the other observations.

Interpretation

The widest portion of the violin represents the region where the score distribution is most concentrated. In this dataset, the distribution is expected to be widest around the **60–70 marks region**.

The relatively narrow ends indicate that fewer students achieved very low scores such as 45–50 or very high scores such as 85.

Therefore, the majority of students have achieved moderate to good scores.

Conclusion

The violin plot provides a detailed view of the exam score distribution. It indicates that the scores are mainly concentrated around **60–70 marks**, with relatively fewer observations at the extreme ends.

Problem 18 – Box Plot – Comparing Two Sections

Introduction

A box plot is used to summarize the distribution of numerical data using the **minimum, first quartile (Q1), median, third quartile (Q3), and maximum**. It is particularly useful for comparing the center and spread of two or more groups.

Here, box plots are used to compare the test scores of **Section A and Section B**.

Given Data

Section A

60, 65, 68, 70, 72, 75, 80

Section B

40, 50, 55, 60, 65, 70, 95

Median Calculation

Section A

The middle value is:

70

Therefore:

$$\text{Median}A=70$$

Section B

The middle value is:

60

Therefore:

$$\text{Median}B=60$$

Range Comparison

Section A

$$80-60=20$$

Range = **20**

Section B

$$95-40=55$$

Range = **55**

Therefore, Section B has substantially greater overall spread.

R Code

```
R
library(ggplot2)

# Create data
section_data <- data.frame(
  Section = c(
    rep("Section A", 7),
    rep("Section B", 7)
  ),
```

```

Score = c(
  60, 65, 68, 70, 72, 75, 80,
  40, 50, 55, 60, 65, 70, 95
)
)

# Calculate medians
median_A <- median(
  section_data$Score[
    section_data$Section == "Section A"
  ]
)

median_B <- median(
  section_data$Score[
    section_data$Section == "Section B"
  ]
)

cat("Median of Section A:", median_A, "\n")
cat("Median of Section B:", median_B, "\n")

# Box plot
ggplot(
  section_data,
  aes(
    x = Section,
    y = Score,
    fill = Section
  )
) +

geom_boxplot(
  width = 0.55,
  color = "black"
) +

geom_jitter(
  width = 0.08,
  color = "red",
  size = 2
) +

labs(
  title = "Box Plot - Section A vs Section B",
  x = "Section",
  y = "Test Score"
) +

theme_minimal() +

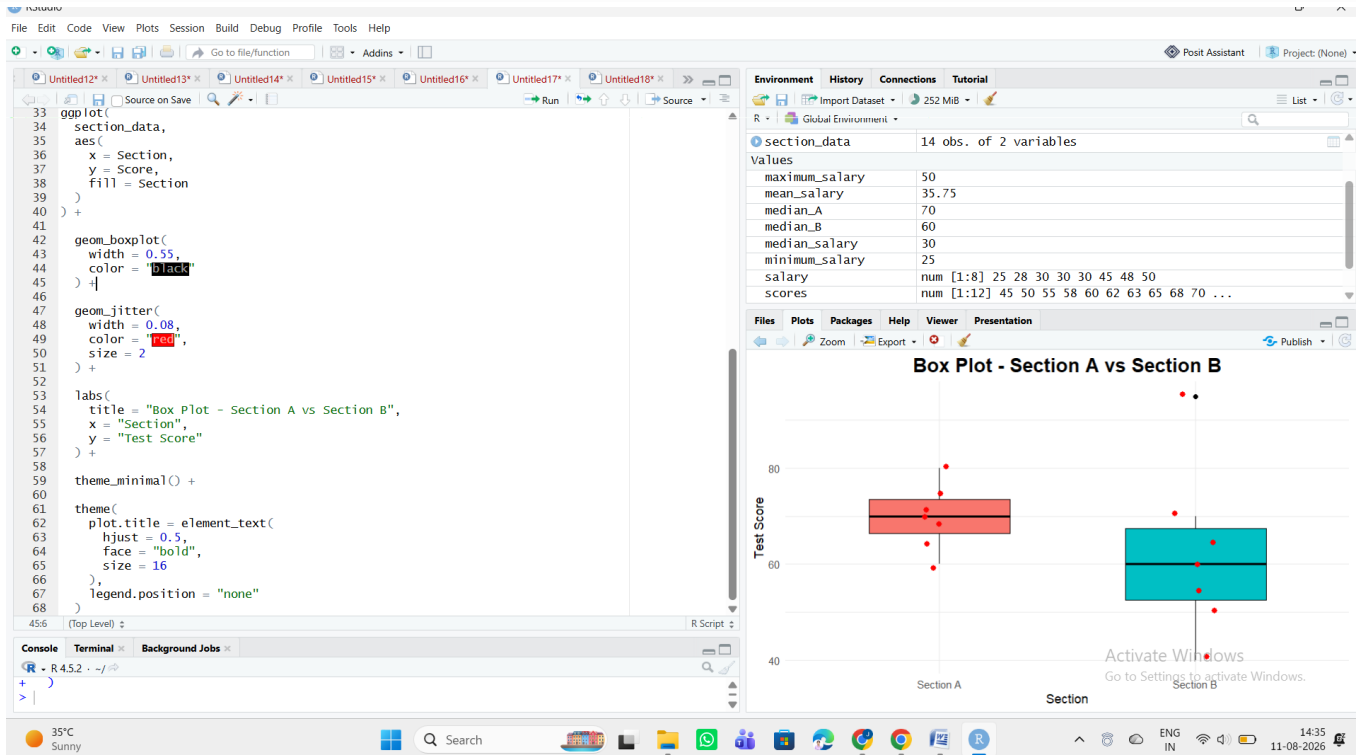
```



```

theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 16
  ),
  legend.position = "none"
)

```



Evidence

Section A has:

- Minimum = 60
- Maximum = 80
- Median = 70
- Range = 20

Section B has:

- Minimum = 40
- Maximum = 95
- Median = 60
- Range = 55

Interpretation

Section A has a **higher median score of 70**, compared with 60 for Section B. This means the typical performance of students in Section A is higher.

Section B has a much wider range of scores, from 40 to 95, compared with 60 to 80 for Section A. Therefore, Section B has much greater variation in student performance.

The score of 95 in Section B extends the upper end considerably, while the score of 40 extends the lower end.

Conclusion

The box plot shows that **Section A has better central performance and more consistent scores**, while **Section B has a lower median and considerably greater variability**.

Therefore, Section A demonstrates more consistent academic performance.

Problem 25 – Stacked Bar Chart – Household Budget

Introduction

A stacked bar chart is used to compare both **total values and the composition of those totals**. Each bar represents a total, while different colored sections within the bar represent individual categories.

In this problem, household spending is compared across two months using Rent, Food, and Other expenses.

Given Data

Month	Rent	Food	Other	Total
Month 1	₹15,000	₹8,000	₹5,000	₹28,000
Month 2	₹15,000	₹9,000	₹6,000	₹30,000

Total for Month 1

$$15000+8000+5000=28000$$

Total for Month 2

$$15000+9000+6000=30000$$

Change in Spending

$$30000-28000=2000$$

Therefore, total spending increased by **₹2,000**.

Percentage increase:

$$\frac{2000}{28000} \times 100 = 7.14\%$$

Thus, household spending increased by approximately **7.14%**.

R Code

```
R
library(ggplot2)

# Create household budget data
budget_data <- data.frame(
  Month = c(
    "Month 1", "Month 1", "Month 1",
    "Month 2", "Month 2", "Month 2"
  ),
  Category = c(
    "Rent", "Food", "Other",
    "Rent", "Food", "Other"
  ),
  Amount = c(
    15000, 8000, 5000,
    15000, 9000, 6000
  )
)

# Calculate totals
total_month1 <- sum(
  budget_data$Amount[
    budget_data$Month == "Month 1"
  ]
)

total_month2 <- sum(
```

```
budget_data$Amount[
  budget_data$Month == "Month 2"
]
```

```
cat(
  "Month 1 Total: ₹",
  total_month1,
  "\n"
)
```

```
cat(
  "Month 2 Total: ₹",
  total_month2,
  "\n"
)
```

```
cat(
  "Increase: ₹",
  total_month2 - total_month1,
  "\n"
)
```

Stacked bar chart

```
ggplot(
  budget_data,
  aes(
    x = Month,
    y = Amount,
    fill = Category
  )
) +
```

```
geom_bar(
  stat = "identity",
  width = 0.6,
  color = "white"
) +
```

```
geom_text(
  aes(
    label = paste0(
      "₹",
      format(
        Amount,
        big.mark = ","
      )
    )
  )
),
```

```
position = position_stack(
  vjust = 0.5
```

```

),
size = 4
) +

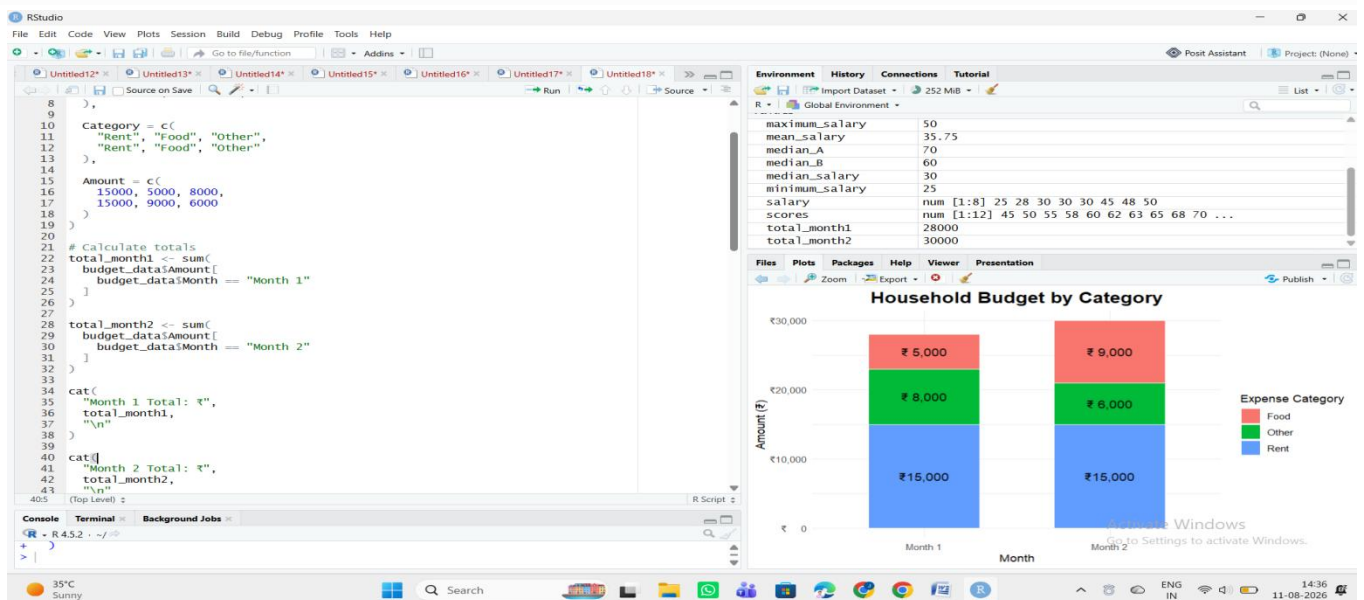
labs(
  title = "Household Budget by Category",
  x = "Month",
  y = "Amount (₹)",
  fill = "Expense Category"
) +

scale_y_continuous(
  labels = function(x) {
    paste0(
      "₹",
      format(
        x,
        big.mark = ",",
        scientific = FALSE
      )
    )
  }
) +

theme_minimal() +

theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 16
  )
)

```



Evidence

The total expenditure in Month 1 is:

₹28,000

The total expenditure in Month 2 is:

₹30,000

Rent remains unchanged at **₹15,000** in both months.

Food increases from **₹8,000 to ₹9,000**, while Other expenses increase from **₹5,000 to ₹6,000**.

Interpretation

The stacked bars show that total household expenditure increased from **₹28,000 to ₹30,000**.

Rent remains the largest component in both months. However, the increases in Food and Other expenses cause the overall household expenditure to rise.

The composition of spending therefore changes slightly between the two months, even though the largest expense category remains Rent.

Conclusion

The stacked bar chart effectively represents both **total household expenditure and category-wise composition**. It shows that spending increased by **₹2,000 (approximately 7.14%)** from Month 1 to Month 2, mainly because of increases in Food and Other expenses.